

Practice 1: Factor pairs

Name: _____ Date: _____

Write every factor pair and explain how you know you found them all.

1. Find every factor pair for 36.

Work space

2. Find every factor pair for 48. How many positive factors does 48 have?

Work space

Practice 2: Listing factors

List factors in order from least to greatest. Check each one by division.

1. List all the positive factors of 24.

Work space

2. List all the positive factors of 54.

Work space

Practice 3: Multiples

Find multiples by multiplying by 1, 2, 3, and so on. Look for the rule in each list.

1. Write the first eight positive multiples of 9.

Work space

2. Is 84 a multiple of 7? Show the multiplication equation that proves your answer.

Work space

Practice 4: Common factors

List factors for both numbers, then identify the overlap.

1. Find all common factors of 18 and 30.

Work space

2. Find all common factors of 28 and 42.

Work space

Practice 5: Greatest common factor

The greatest common factor (GCF) is the largest factor shared by two or more numbers.

1. Find the GCF of 24 and 36. Show at least two methods.

Work space

2. A teacher has 45 pencils and 60 erasers. What is the greatest number of identical supply kits she can make? How many pencils and erasers go in each kit?

Work space

Practice 6: Divisibility by 2, 5, and 10

Use the last digit. A number divisible by 2 is even; one divisible by 5 ends in 0 or 5; one divisible by 10 ends in 0.

1. For each number, list which of 2, 5, and 10 divide it: 1,240; 1,275; 1,286.

Work space

2. Write the least four-digit number divisible by both 2 and 5. Explain your reasoning.

Work space

Practice 7: Divisibility by 3 and 9

Add the digits. If the digit sum is divisible by 3 or 9, the original number is divisible by that number.

1. Is 4,572 divisible by 3, by 9, by both, or by neither? Prove it with the digit sum.

Work space

2. Replace the box in 63[] so the number is divisible by 9. Find every possible digit.

Work space

Practice 8: Divisibility by 4 and 8

For 4, check the last two digits. For 8, check the last three digits.

1. Determine whether 7,316 is divisible by 4. Show the test you used.

Work space

2. Determine whether 5,248 is divisible by 8. Show the test you used.

Work space

Practice 9: Prime and composite numbers

A prime has exactly two positive factors. A composite number has more than two. The number 1 is neither.

1. Classify each number as prime, composite, or neither: 1, 17, 21, 29, 35.

Work space

2. Find all the prime numbers between 40 and 60. Explain how you ruled out composites.

Work space

Practice 10: Prime factorization

Prime factorization writes a composite number as a product of prime numbers.

1. Write the prime factorization of 72.

Work space

2. Write the prime factorization of 210. Check by multiplying the prime factors.

Work space

Practice 11: Factor trees

A factor tree breaks a composite number into factors until every branch ends in a prime.

1. Draw a factor tree for 84 and write its prime factorization.

Work space

2. Draw a different factor tree for 90. Explain why its prime factorization matches any other correct tree.

Work space

Practice 12: Least common multiple

The least common multiple (LCM) is the smallest positive multiple shared by two or more numbers.

1. Find the LCM of 6 and 8 by listing multiples.

Work space

2. Find the LCM of 12 and 15 using prime factors or a list.

Work space

Practice 13: Common multiples in context

Use common multiples when repeating events must happen together.

1. A red light flashes every 4 seconds and a blue light every 6 seconds. If they flash together now, after how many seconds will they flash together again?

Work space

2. One bus arrives every 10 minutes and another every 15 minutes. They arrive together at 8:00 a.m. When is the next simultaneous arrival?

Work space

Practice 14: Arrays and area

An array shows equal rows and columns. Its total is a product, so its dimensions are factor pairs.

1. Draw two different rectangular arrays with 24 tiles. Label the rows and columns.

Work space

2. A rectangle has an area of 48 square units and whole-number side lengths. List every possible pair of side lengths.

Work space

Practice 15: Missing factors

Use inverse operations and factor knowledge to find an unknown factor.

1. Find the missing factor: $9 \times [] = 117$. Explain how you checked.

Work space

2. A number has factors 1, 2, 4, 8, 16, and 32. What is the number? Explain why no other factor is needed in the list.

Work space

Practice 16: Number-property puzzles

Use several tests together. Show every condition you use and check your final answer.

1. Find every number from 100 to 130 that is divisible by both 3 and 5.

Work space

2. I am a two-digit composite number, divisible by 4, and my digits add to 9. What numbers could I be?

Work space

Practice 17: Grouping with factors

Use the GCF to make equal groups with no items left over.

1. A club has 32 students and 48 parents. What is the greatest number of identical teams possible? How many students and parents are on each team?

Work space

2. A baker has 72 chocolate and 90 vanilla cookies. What is the greatest number of identical gift boxes she can make? How many of each flavor per box?

Work space

Practice 18: Remainders and divisors

In a division statement, the remainder must be less than the divisor. Use that rule to interpret a situation.

1. Find the quotient and remainder for $157 \div 12$. Check with $\text{divisor} \times \text{quotient} + \text{remainder}$.

Work space

2. A camp has 98 children. Vans hold 8 children each. How many vans are needed, and why can't the answer be only the quotient?

Work space

Practice 19: Mixed review

Choose a strategy and write enough work for another reader to follow.

1. Find the GCF and LCM of 18 and 24. Describe how the two answers are related to the factor lists.

Work space

2. Is 1,008 divisible by 2, 3, 4, 5, 8, 9, or 10? List every true statement and prove each with a divisibility rule.

Work space

Practice 20: Challenge and explain

Use factorization, divisibility, and clear mathematical language.

1. Find the smallest positive number with exactly three distinct prime factors and a total of 24 positive factors. Show your reasoning.

Work space

2. A student says, “Every even number is composite.” Is the statement always true? Give a counterexample or prove it, then correct the statement if needed.

Work space